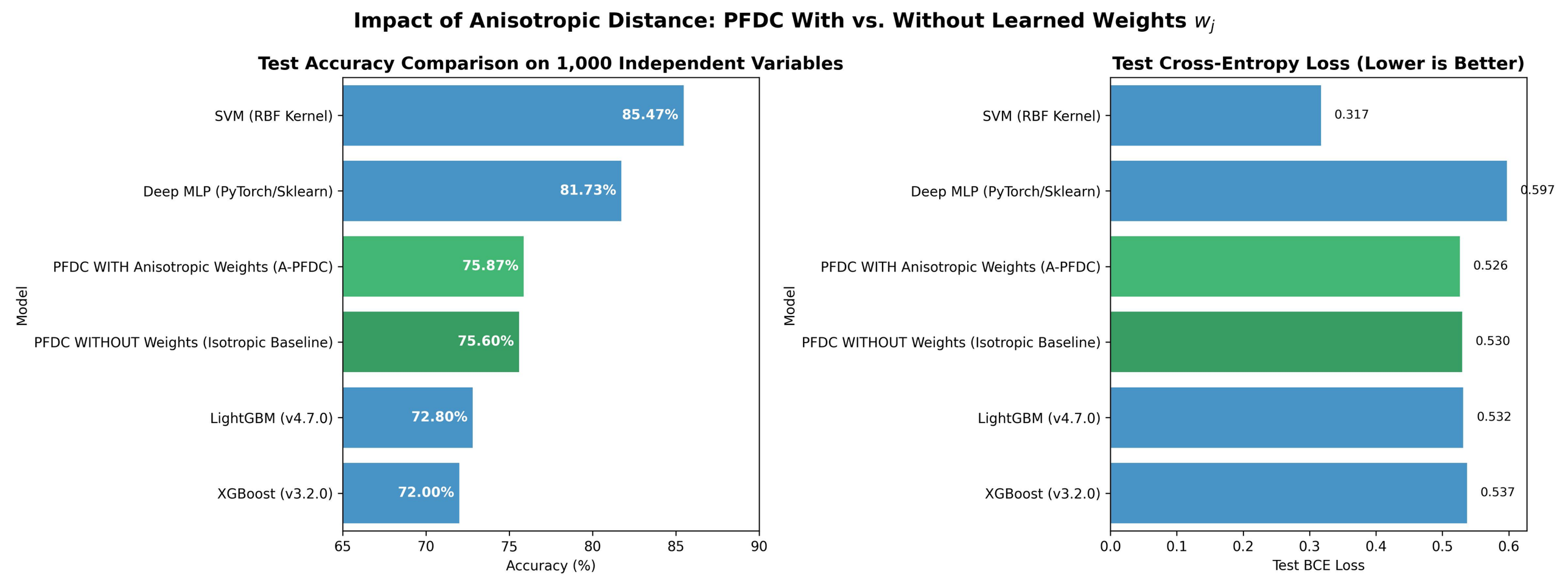

Impact of Anisotropic Distance: PFDC With vs. Without Learned Weights w_j
Test Accuracy Comparison on 1,000 Independent Variables
SVM (RBF Kernel)
Deep MLP (PyTorch/Sklearn)
PFDC WITH Anisotropic Weights (A-PFDC)
PFDC WITHOUT Weights (Isotropic Baseline)
LightGBM (v4.7.0)
XGBoost (v3.2.0)
85.47%
81.73%
75.87%
75.60%
72.80%
72.00%
Model
65
70
75
80
85
90
Accuracy (%)
Test Cross-Entropy Loss (Lower is Better)
SVM (RBF Kernel)
Deep MLP (PyTorch/Sklearn)
PFDC WITH Anisotropic Weights (A-PFDC)
PFDC WITHOUT Weights (Isotropic Baseline)
LightGBM (v4.7.0)
XGBoost (v3.2.0)
0.317
0.597
0.526
0.530
0.532
0.537
Model
0.0
0.1
0.2
0.3
0.4
0.5
0.6
Test BCE Loss